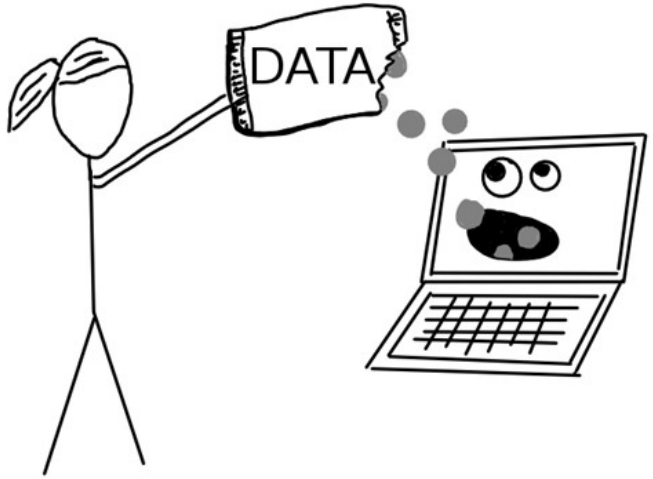

Enset DE Mohammedia- Master
II-BDCC

Année universitaire 2023/2024

MLOPS – Research and Developpment Environment

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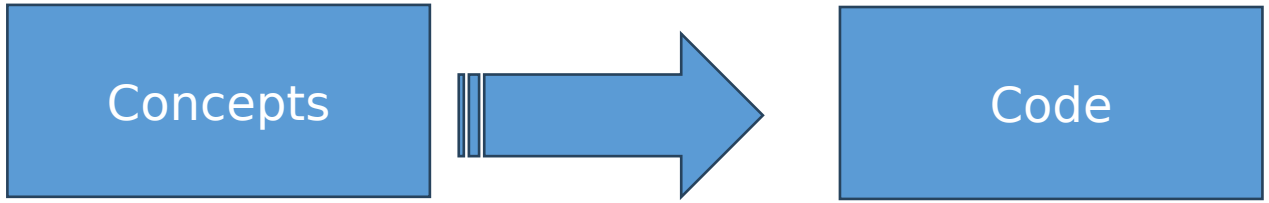
how to learn machine learning well

If you can’t implement it, then you don’t understand it

How to practice

- Recall: this is a hands-on course
- Get your hands dirty!



- **Remember: a real job / project involves actually writing code**

MLOPS- Research and Developpment Outlines

1. Data Analysis pipline
2. Feature Engineering pipeline
3. Feature Selection pipeline
4. Model Training pipeline
5. Scoring pipeline

Machine Learning Project end to end

1. Data set

Machine Learning Project end to end

2. **Data Preprocessing:** handling missing values, dealing with outliers, standardizing or normalizing features, encoding categorical variables, and other data transformations to ensure that the data is suitable for modeling.

- Allows to an algorithm ML the ability of interpreting easily the data's features

3. **Exploration and Data Analysis (EDA):** examining and analyzing the dataset to gain insights and a better understanding of its characteristics.

- Visualizing data, calculating statistics, identifying patterns, and exploring relationships within the dataset
- Inform feature selection, model choice, and data preprocessing decisions

4. **Feature Engineering**: creating, selecting, or transforming input features (characteristics) to improve a model's performance.

- Designing relevant and informative input data for machine learning algorithms
- And then possibly have a model more accurate and precise in predictions

Machine Learning Project end to end

5. Model Selection: this step involves choosing the appropriate machine learning algorithm or model architecture based on the nature of the task and dataset.

- This step is critical for achieving optimal model performance and accuracy.

6. Model Training: involves using a selected algorithm to fit the model to the training data, adjusting its parameters to minimize a specified loss function.

- This process allows the model to learn patterns and relationships in the data and become capable of making predictions

7. Model Evaluation: assesses the performance of a trained model using independent test or validation data

- This step helps determine how well the model generalizes to unseen data
- For Classification Task: Measuring its accuracy, precision, recall, F1 score, or other relevant metrics .
- For Regression Task: Measuring its MSE, RMSE, or other relevant metrics.

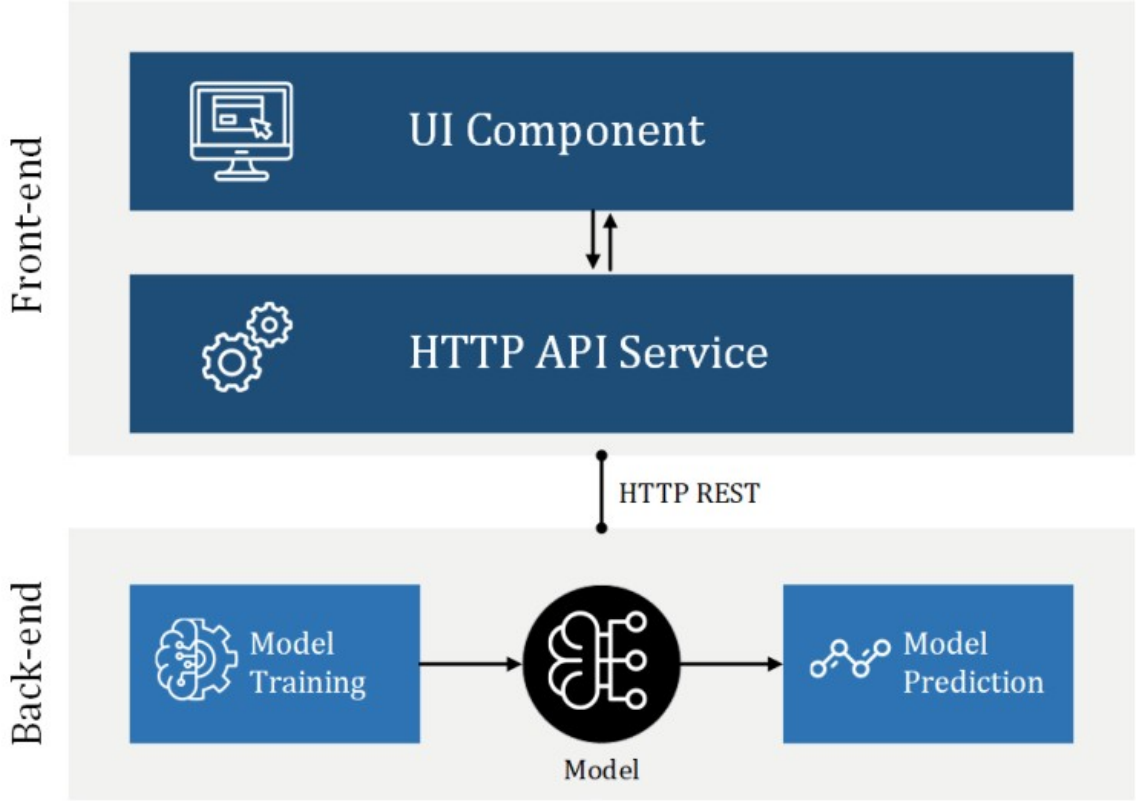
8. Model Optimization: involves fine-tuning the model's hyperparameters, architecture, and training process to improve its performance, typically based on evaluation results.

- This iterative process aims to enhance the model's predictive power and generalization capabilities for better results on unseen data.

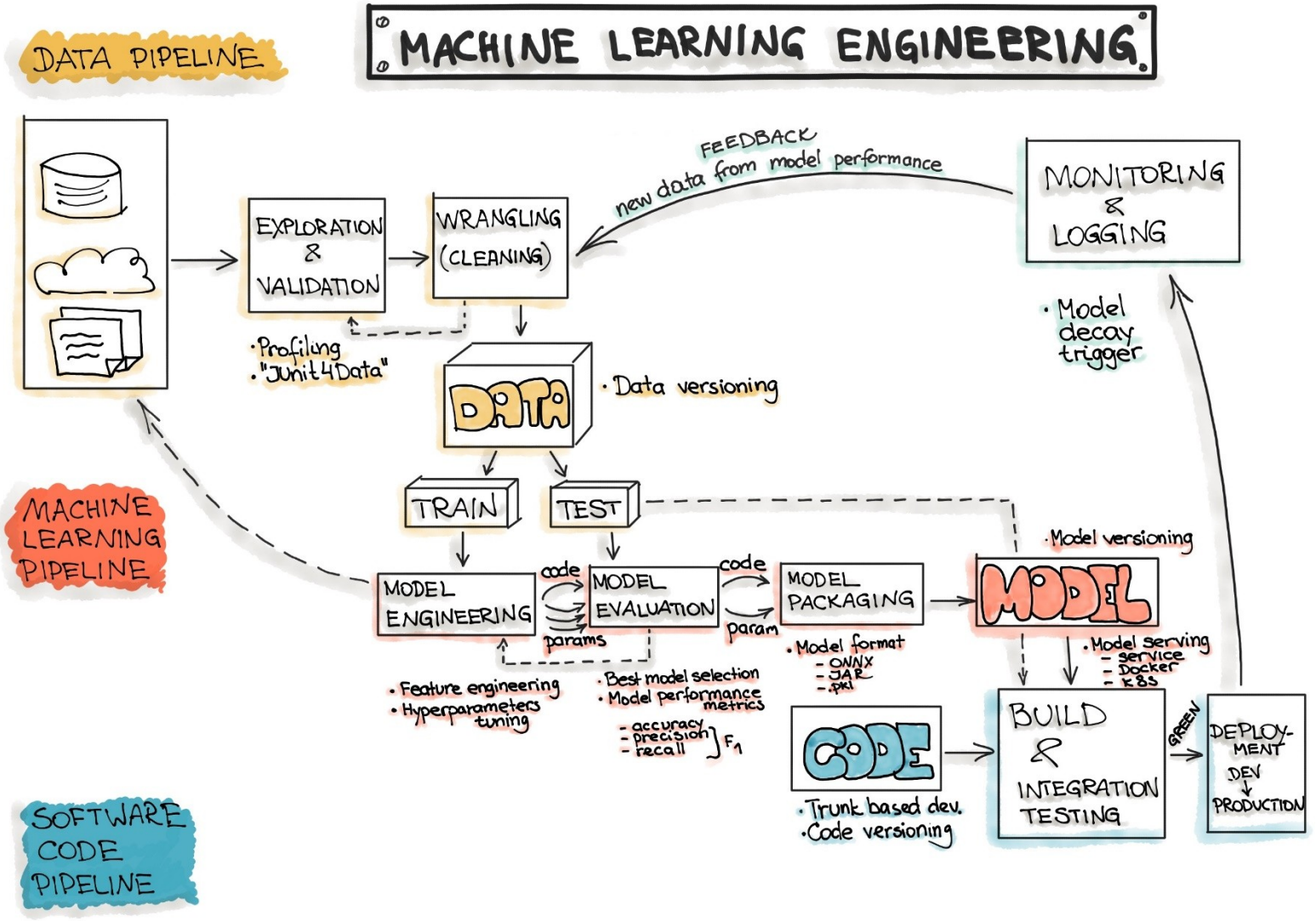
Machine Learning Project end to end

9. Model Deployment: involves integrating the trained model into a production environment or system, making it available for real-time predictions or decisions.

- This step ensures that the model can be used to address the intended problem in practical applications
- Involves creating APIs or software interfaces for its seamless integration.



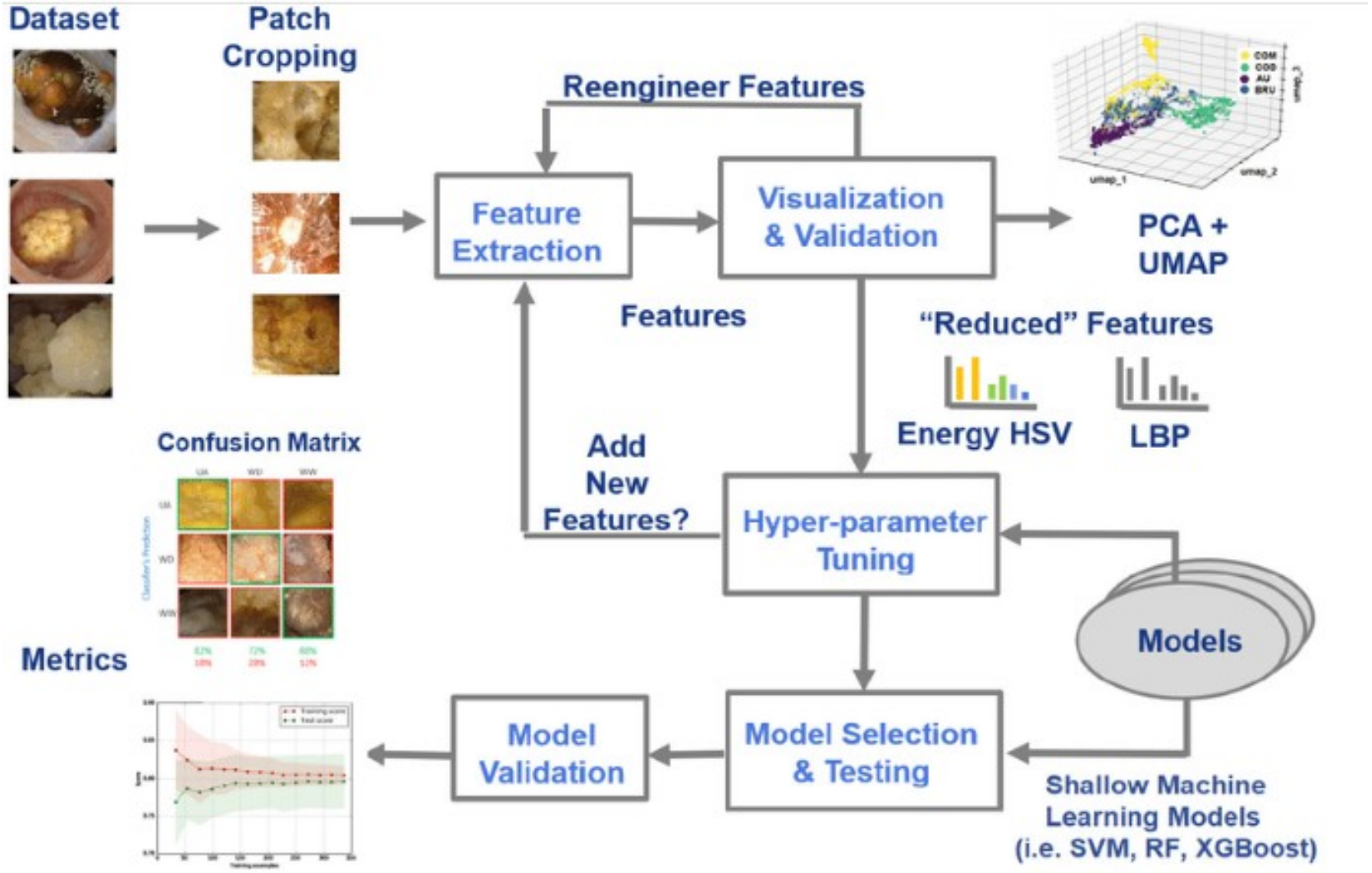
Machine Learning Project end to end workflow



Example 1

https://ml-ops.org/content/end-to-end-ml-workflow

Machine Learning Project end to end workflow



Example 2

https://www.researchgate.net/figure/Illustration-of-the-training-of-the-six-shallow-machine-learning-models-feature_fig3_358144402

Application
Prediction Sale Price of Houses

how to carry out the project?

work on two different environments

Research and Development Environment

- Represents a dedicated space and infrastructure where researchers and data scientists can experiment, innovate, and explore new techniques and approaches to advance the project's goals.
- Use notebook to

Production Environment

- Represents the operational and real-world deployment phase where the trained machine learning model is integrated into a system or application to provide predictions.
  - Librairies
  - Configurations
  - .py files
  - ...

- **Observe the structure of the notebook file**

Import dependencies

Local training data

Prep train/ test data

Config

Data preprocessing Pipeline

Train model

Evaluate the model

Persist the model

Score new data

Research Environment/ Jupyter notebook code

https://scikit-learn.org/stable/modules/preprocessing.html

Data set

- Objective: predict SalePrice of houses
- house-prices dataset
- **Download the data set from kaggle:** https://www.kaggle.com/c/house-prices-advanced-regression-techniques/data
  - Download the train.csv & test.csv
- Explore rows and columns
  - Shape: (1460, 81)
  - 79 predictive variables and 1 target variable

Data set- columns

Predictible variable			
~~id~~			
1. MSSubClass	21.RoofStyle	41.CentralAir	61.GarageCars
2. MSZoning	22.RoofMatl	42.Electrical	62.GarageArea
3. LotFrontage	23.Exterior1st	43.1stFlrSF	63.GarageQual
4. LotArea	24.Exterior2nd	44.2ndFlrSF	64.GarageCond
5. Street	25.MasVnrType	45.LowQualFinSF	65.PavedDrive
6. Alley	26.MasVnrArea	46.GrLivArea	66.WoodDeckSF
7. LotShape	27.ExterQual	47.BsmtFullBath	67.OpenPorchSF
8. LandContour	28.ExterCond	48.BsmtHalfBath	68.EnclosedPorch
9. Utilities	29.Foundation	49.FullBath	69.3SsnPorch
10.LotConfig	30.BsmtQual	50.HalfBath	70.ScreenPorch
11.LandSlope	31.BsmtCond	51.BedroomAbvGr	71.PoolArea
12.Neighborhood	32.BsmtExposure	52.KitchenAbvGr	72.PoolQC
13.Condition1	33.BsmtFinType1	53.KitchenQual	73.Fence
14.Condition2	34.BsmtFinSF1	54.TotRmsAbvGrd	74.MiscFeature
15.BldgType	35.BsmtFinType2	55.Functional	75.MiscVal
16.HouseStyle	36.BsmtFinSF2	56.Fireplaces	76.MoSold
17.OverallQual	37.BsmtUnfSF	57.FireplaceQu	77.YrSold
18.OverallCond	38.TotalBsmtSF	58.GarageType	78.SaleType
19.YearBuilt	39.Heating	59.GarageYrBlt	79.SaleCondition
20.YearRemodAd	40.HeatingQC		

Target variable
1. SalePrice

- 44 Categorical variables
- 35 numerical variables
- Target is numeric

Part 1
Data Analysis pipeline

Data Analysis pipeline

In this part, we will analyse the following:

1. The target variable
2. Variable types (categorical and numerical)
3. Missing data
4. Numerical variables
    - Discrete
    - Continuous
        - Distributions
        - Transformations
5. Categorical variables
    - Cardinality
    - Rare Labels
    - Special mappings

Data Analysis pipeline

1. **The target variable**
2. Variable types (categorical and numerical)
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1. Target variable analysis

- What we need to knwo:
  - The type of the target variable (regression or classification)
  - Continus target
    - Distribution
    - Improve the spread of the target by appliying mathematical techniques: log transformation, power transformation, normalization, centering, custom transformation, scale change
  - Discrete: log transformation, power transformation, normalization, centering, custom transformation, ordinal Encoding, One-Hot encoding,
  - Categorial: One-Hot encoding, **Label Encoding (in case of ordered values), Frequency Encoding, Embedding**
    - **Embedding:** represents categorical data as continuous vectors, enabling models to capture relationships and similarities between categories
  - binary

Data Analysis pipeline

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2. Variables types

- What we need to knwo:
  - identify categorical variables
  - identify numerical variables

Data Analysis pipeline

1. **The target variable**
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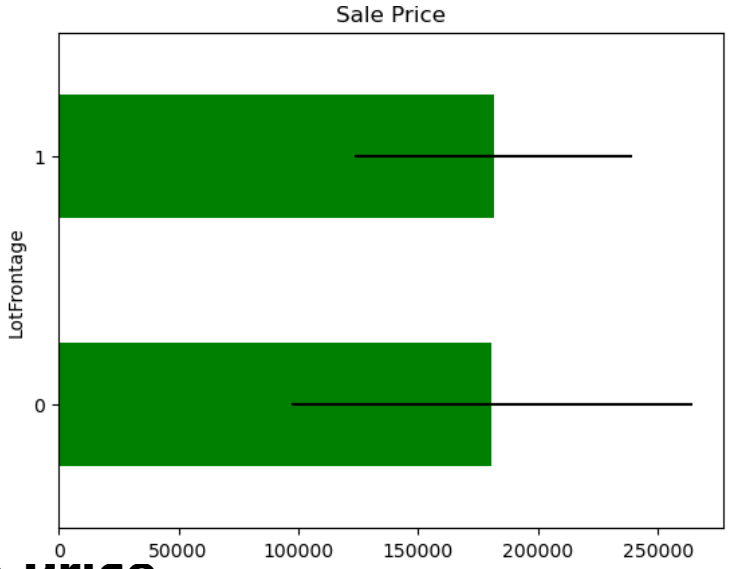
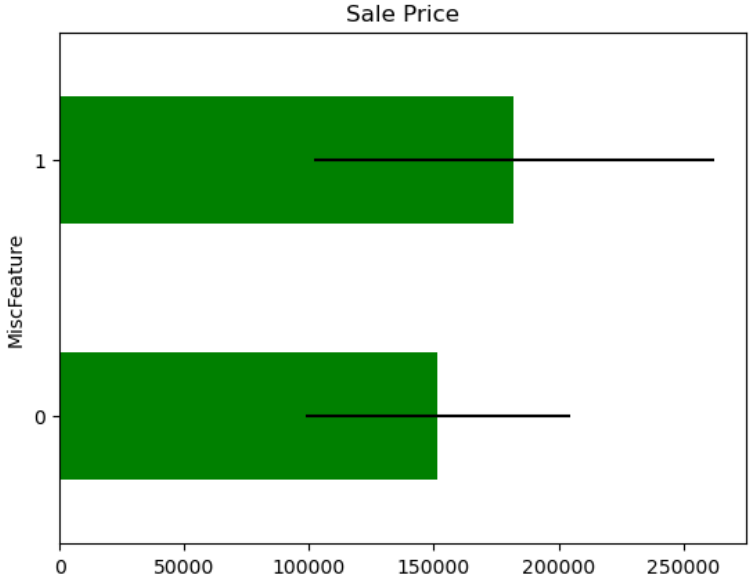
3. Missing Data

- What we need to do:
  - How many variables has Nan values
  - Which variables have Nan
  - Which categorical variables have Nan
  - Which numericla variables have Nan
  - **what is the percentage of null values for each variable**
  - **What is the relationship between missing data and Sale Price**
    - **how depends the output to variables with missing v**
  - What Numerical Variables we have in the dataset
  - Which information can we have form temporal variables
  - Find which variables are discrete and how can they influence t
  - Find which variables are continuous and verify their distributio transformation can be made



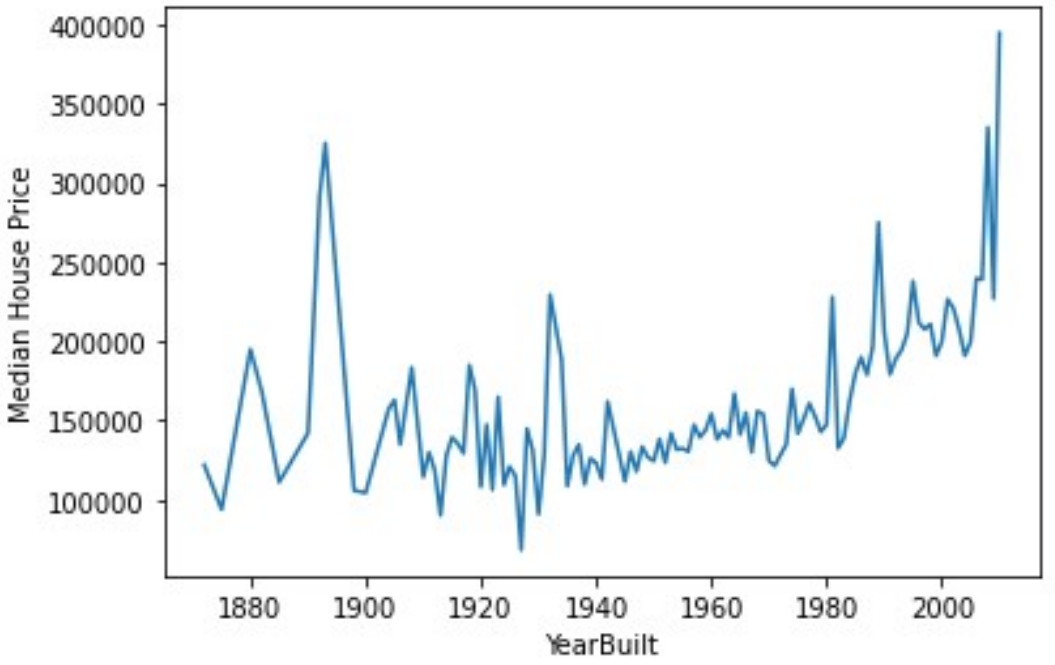


Data Analysis pipeline

1. **The target variable**
2. **Variable types (categorical and numerical)**
3. **Missing data**
4. **Numerical variables**
    - **Discrete**
    - **Continuous**
    - **Distributions**
    - **Transformations**
5. Categorical variables
    - Cardinality
    - Rare Labels
    - Special mappings

4. Numerical Data

- What we need to do:
  - Which Numerical Variables we have in the dataset
  - Find which variables are temporal and how can they influence the price and which information can we have form them
  - Find which variables are discrete and how can they influence the price
  - Find which variables are continuous and verify their distribution, and which transformation can be made for each continuous variable
  - Find skewed variables and see how predictive they



Data Analysis pipeline

1. The target variable
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    - Discrete
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    - Rare Labels
    - Special mappings

5. Categorical Variables

- What we need to do:
  - Find skewed (asymmetric) variables and see how predictive they are
  - How many different categories are present in each categorical variable
  - Find which categorical variable are ordered and how can be encoded
  - Which labels are under-represented in the dataset and why we shood remove or replace them
  - Find the relationship between the categories of the different variable and the house sale price

5. Categorical Variables

Machine Learning Techniques & Categorical Data

- Incode categorical data into numerical data before use them is a crucial step before modeling step
- Some techniques can not work with categorical data directly: neural networks, knn, logistic regression, linear regression…
  - These techniques require transformations
- Some techniques can work with categorical data directly: decision trees, random forest
  - But, Some implementations of machine learning algorithms require all data to be numerical like sklearn implementation

5. Categorical Variables

Ordinal Encoding for Oredered Categorical Data

- Ordinal encoding
  - For example: size (XS,S,M,L,XL,XXL), the order have a sense
  - machine learning algorithms may be able to understand and harness this sense.

Example

```
import pandas as pnd

data = ['M','XL','M','XS','S','XS','S']

df = pnd.DataFrame(data, columns=['Size'])

scale_mapper = {"XS":1, "S":2, "M":3,"L":4,"XL":5}

df["Size"] = df["Size"].replace(scale_mapper)
```

5. Categorical Variables

One-Hot Encoding for Non Ordrered Categorical Data

- One-Hot Encoding:
  - For example: color (red,green,blue), the order not make sense

Example

```
from numpy import asarray
from sklearn.preprocessing import OneHotEncoder
# define data
data = asarray([['red'], ['green'], ['blue']])
print(data)
print("--------------------------------")
# define one hot encoding
encoder = OneHotEncoder(sparse=False)
# transform data
onehot = encoder.fit_transform(data)
print(onehot)
```

Part 2
Features Engineering pipeline

Features Engineering pipeline

1. Separate data into train and test
   - How this separation allows us to avoid over-fitting
   - Fix the seed
2. Deal with missing data
   - For categorical variables
   - For numerical variables

3. Transform data
   - Categorical variables
   - Numerical variables
   - Temporal variables
   - Discrete variables
4. Deal with rare lables
5. Encode categoricla variable
6. Scale continuous variables
7. Save train and test data in new files

Part 3
Features Selection pipeline

Features Selection pipeline

- A feature selection pipeline represents a systematic and automated process for selecting the **most relevant features** (input variables) from a larger set of features in a machine learning project.
- This pipeline is a series of steps or components that are used to preprocess and choose the most important features **before training a machine learning model**.
- Feature selection is essential for **improving model performance**, reducing overfitting, and enhancing interpretability.
- **Different feature selection algorithms**, such as **Recursive Feature Elimination** (RFE), **L1 regularization** (Lasso), or forward/backward selection, are applied to choose the top-performing features based on their scores.

Part 4
Model Training pipeline

Model Training pipeline

- Model Selection
- Model Training
- Model Tuning

Part 5
Scoring pipeline

Scoring pipeline

- Test the model on data never seen